E-Tendering System: Manual Testing Guide

Step-by-Step Walkthrough, Scoring Rubric & Verification Case Study

Research Demonstration Case Study

Case Study: Procurement of Building Materials for New Engineering Complex
Tender Title: Procurement of High-Grade Structural Steel & Reinforced Concrete for University Engineering Complex
Procuring Entity: Ministry of Higher Education & Infrastructure Works
Evaluation Weights: Technical Evaluation = 70% | Financial Evaluation = 30%
Competing Contractors: Alpha Buildtech Ltd (Bidder 1) vs Beta Construction Corp (Bidder 2)

Stage 1: Tender Advertisement (Procurement Officer)

1. In the top-right account dropdown, select 'Procurement Officer / Admin (Acc #0)'.
2. Navigate to the 'Officer Dashboard' tab.
3. Click 'Advertise New Tender' button.
4. Enter Title: 'Procurement of High-Grade Structural Steel & Concrete (TND-2026-ENG-001)'.
5. Set Submission Window: '60 minutes', Technical Weight: '70%', Financial Weight: '30%'.
6. Click 'Advertise Tender On-Chain'. The smart contract initializes the tender in state 'Advertised' and records weights.

Stage 2: Prequalification Window & Officer Approval

1. In Officer Dashboard, click 'Open Prequalification Window'. State transitions to 'PrequalificationOpen'.
2. Switch to 'Bidder 1: Alpha Engineering Ltd (Acc #1)' in the account dropdown.
3. Navigate to the 'Bidder Portal' tab and click 'Register for Prequalification'.
4. Switch back to Officer (Acc #0) in Officer Dashboard. Locate Alpha Engineering under Registered Bidders and click 'Approve Prequalification'.
5. (Optional) Repeat for Bidder 2 (Beta Solutions Corp) to create competitive scoring.

Stage 3: Proposal Submission & Cryptographic Hashing

1. As Officer (Acc #0), click 'Open Submission Window'. State transitions to 'Submitted'.
2. Switch to Bidder 1 (Acc #1) and go to 'Bidder Portal'.
3. Under 'Upload & Submit Sealed Bid Proposal', select the sample proposal document (or click 'Generate Sample Proposal File').
4. Observe that the client computes the Keccak-256 hash in browser memory before uploading.
5. Click 'Submit Hashed Bid On-Chain'. The smart contract records the document hash and timestamp.

Stage 4: Tender Opening & Blockchain Fast-Forward

1. Switch to Officer (Acc #0) on the Officer Dashboard.
2. Notice that if you attempt to open the tender before the deadline, the contract blocks the transaction.
3. Click the amber button: 'Fast-Forward Blockchain Time Past Deadline' (advances the local EVM clock).
4. Click 'Open Tender Bids'. State transitions to 'Opened'. Bids are now officially unsealed for grading.

Smart Contract Weighted Scoring Formula

Total Score = (Technical Score × Weight_tech) + (Financial Score × Weight_fin) / 100

Stage 5: Evaluation & Mathematical Scoring Criteria

Grading Criteria Rubric for Building Materials Procurement:
• Technical Evaluation (0 - 100 pts / Weight: 70%):
  - Compliance with Material Specifications (ASTM A992 Steel, Grade 60 Rebar, Portland Type I): 40 pts
  - Quality Assurance Certifications (ISO 9001:2015, Mill Test Certificates): 25 pts
  - Delivery Logistics, Staging Plan & Factory Lead Time (under 45 days): 20 pts
  - Contractor Track Record in Major Civil Infrastructure: 15 pts

• Financial Evaluation (0 - 100 pts / Weight: 30%):
  - Competitive Total Bid Price relative to Government Baseline Estimate: 60 pts
  - Payment Milestones & Retainage Terms: 25 pts
  - Price Validity Guarantee (90 days fixed): 15 pts

Example Evaluation Execution:
• Enter Technical Score: 92 | Financial Score: 88 for Alpha Buildtech.
• Smart Contract Calculation: (92 × 70 + 88 × 30) / 100 = 6440 + 2640 = 90.80 Total Score.
• Click 'Submit Evaluation On-Chain'. Note: Scores are write-once; re-evaluations revert to prevent tampering.

Stage 6: Automated On-Chain Tender Award

1. As Officer (Acc #0), click 'Award Tender'.
2. The smart contract iterates over all submitted bidders, verifies that all bids have been graded, and automatically sets the winner to the contractor with the highest total score.
3. The winning address and score are emitted in event 'TenderAwarded'.

Stage 7: Final Contract Archiving & Permanent Seal

1. Under Stage 7, upload the executed Final Contract Agreement or click 'Generate Sample Final Contract File'.
2. Click 'Seal & Archive Tender Record On-Chain'.
3. The tender is now permanently locked in state 'Archived'. All subsequent write operations are rejected by the smart contract.

Public Audit Trail Verification (All Roles)

1. Navigate to the 'Audit Trail' tab in the main navigation.
2. Select the Tender ID from the selector.
3. Review the complete chronological timeline from 'Advertised' to 'Archived'.
4. Verify that each event displays its exact block timestamp, calling wallet address, transaction hash, and document fingerprints.
5. Confirm that the winner and winning score match the on-chain evaluation calculation.